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27 March 1995

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21 July 1988

# MILITARY SPECIFICATION

## BATTERY RELAY CONTROL UNIT, AIRCRAFT

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1 SCOPE

1.1 Scope This specification covers the requirements for the design, performance, manufacture, and acceptance of a battery relay control unit (BRCU) for use in aircraft to prevent unintentional discharge of aircraft batteries.

## 2 APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)

SPECIFICATIONS

MILITARY

| | | |
|---|---|---|
| MIL-P-116 | - | Preservation, Methods of |
| MIL-P-15024 | - | Plates, Tags and Bands for Identification of Equipment |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Code 6092, Bldg 2949, Naval Surface Warfare Center Crane Division, 300 Highway 361, Crane, Indiana 47522-5001, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 5945

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DOD-C-38999/42 - Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, and Breech Coupling), Environmental Resistant, Removable Crimp and Hermetic Solder Contacts, Receptacle, Box Mounting Flange, Breech Coupling, Removable Crimp Contacts Series IV, Metric

DOD-C-38999/46 - Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, and Breech Coupling), Environmental Resistant, Removable Crimp and Hermetic Solder Contacts, Plug,Breech Coupling, EMI Grounding, Removable Crimp Contacts, Series IV, Metric

MIL-I-45208 - Inspection System Requirements

MIL-I-81023 - Inductor, 28 V.D.C. Laboratory Test, General Specification for

MIL-E-81910 - Electric Power Generating and Control Equipment, Aircraft, Testing, General Specification for

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of U.S Military Property

MIL-STD-202 - Test Methods for Electronic and Electric Component Parts

MIL-STD-454 - Standard General Requirements for Electronic Equipment

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of

MIL-STD-704 - Aircraft Electric Power Characteristics

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

MIL-STD-889 - Dissimilar Metals

MIL-STD-2073-1 - DOD Material Procedures for Development and Application of Packaging Requirements (Part 1 of 2 parts)

(Unless otherwise indicated, copies of the federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099)

2 1.2 Other Government document The following other Government document forms a part of this document to the extent specified herein. Unless otherwise specified, the issue is that cited in the solicitation

SD-6 - Provisions Governing Qualification (Qualified Products List)

(Copies of SD-6 are available from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099 )



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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Qualification BRCUs furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4 4 and 6.3).

3.2 Materials.

3 2 1 Threaded parts. All internal or external parts which are threaded shall be positively locked Accidental loosening of threaded elements shall be prevented by self-locking nuts, safety wiring, or similar methods appropriate for application in the BRCU.

3.2 2 Fungus Materials used shall be fungus inert.

3 2 3 Toxic. Materials which produce toxic effects under any conditions, including fire, shall not be incorporated in the design of the BRCU

3 2 4 Flammable. Materials which will support combustion shall not be used Definitions and restrictions on the use of flammable materials shall be in accordance with MIL-STD-454, Requirement 3.

3 2 5 Metals Metals shall be of a corrosion-resisting type or suitably processed to resist corrosion

3 2.6 Dissimilar metals. The use of dissimilar metals in contact with each other (particularly brass, copper, or steel in contact with aluminum or other metals) shall be avoided. Where contact between dissimilar metals is unavoidable, the contacting metals shall be suitably protected against electrolytic corrosion When protection is used, it shall be a type that offers a low impedance path to high frequency current Dissimilar metals are defined in MIL-STD-889

3 3 Design and construction.

3 3 1 Cooling The BRCU shall be convection cooled.

3.3.2 Dimensions The dimensions shall be in accordance with figure 1. Where only maximum or minimum dimensions are shown, the part need not have the shape shown, but the part, including all protrusions, shall be contained within the outline shown.

3 3 3 Weight. The weight shall be not greater than 0.25 kilogram.

3.3.4 Mounting. The BRCU shall be capable of being mounted in any position. Mounting provisions shall be as shown on figure 1.

3.3.5 Color. The BRCU shall be finished in 17875 white in accordance with FED-STD-595. The finish shall be unaffected by testing required by this specification



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3.3 6 Electrical connector The BRCU shall be provided with a D38999/42WB5PN connector in accordance with DOD-C-38999/42 as shown on figure 1. The connector shall be oriented as shown on figure 1 with the main key on the same side of the unit as the label The equipment structure shall not be used as a current path except for shielding against electromagnetic interference The connector mounting flange shall be secured to the BRCU by hardware designed to impede removal

3.3 7 External adjustments The BRCU shall have no external adjustments

3.3 8 Input voltage The BRCU shall be capable of withstanding voltages in the range of 0 to 33 volts direct current (VDC) from pin A to pin B The load source voltage will be at the same potential as the voltage from pin A to pin B The unit shall be capable of initiating a ground path from pin C to pin E when voltage in the range of 18 to 33 VDC is applied, and capable of maintaining and terminating the ground path when voltage in the range of 10 to 33 VDC is applied The ground path shall not exhibit any uncontrolled on-off cycling, oscillation, or other evidence of instability when the unit is operated within the specified input voltage range

3 3 9 Input current The input current required to operate the BRCU shall be not greater than 0 050 ampere None of the load current input at connector pin C shall be used to operate the BRCU

3 3 10 Input power characteristics The BRCU shall meet the requirements for utilization equipment specified in MIL-STD-704 except as otherwise specified herein

3 3 11 Duty rating. The BRCU shall be designed for continuous duty

3 3 12 Maintainability and repairability The BRCU shall require no maintenance to meet its requirements The unit shall be constructed to hinder disassembly or access to internal components The unit shall contain no internal provisions, such as connectors, for repair or maintenance Any visible hardware items shall be of a type designed to impede removal.

3 3 13 Electrical connection sequence The BRCU shall be designed such that any sequence of application or removal of electrical inputs at the connector pins shall not damage the BRCU

3 3 14 Insulation resistance When tested as specified in 4 8 6, the BRCU insulation resistance shall be not less than 100 megohms

3.4 Performance

3 4 1 Electrical performance The BRCU shall meet the performance requirements specified in 3 4 1 1 through 3 4 1 7 when connected as specified on figure 2 and when input and load source voltage in the range specified by 3.3 8 is applied from pin A to pin B

3 4.1 1 Ground path The BRCU pin C shall provide a ground path when ground is applied to pins D and E and the input voltage is between 18 and 33 VDC. The ground path provided by pin C shall be a current path to pin E, this path shall be capable of conducting 0 75 ampere with a resulting voltage drop not greater than 100 millivolts The ground path shall be free of abnormal interruptions and transients The BRCU shall not be required to initiate the ground path when input operating voltage is below 18 0 VDC If no ground exists at pin E, pin C shall be disconnected from ground, and current shall be not greater than 10 microamperes through any pin



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3 4.1.2 Timing period. If pins D and E are grounded such that pin C is providing a ground path as defined by 3.4.1.1 and the ground is removed from pin D, pin C shall continue to provide a ground path in a timed mode for 5.0 ± 0.5 minutes after the removal of the ground at pin D. The ground path shall be free of abnormal interruptions and transients. Following the timing mode, the ground path from pin C to pin E shall be disconnected, and current shall be not greater than 0 005 ampere through any pin If ground is reapplied to pin D during the 5-minute timing mode for 100 milliseconds or longer, the timer shall reset and provide a ground path at pin C for 5.0 ± 0 5 minutes following removal of the ground at pin D

3 4 1 3 Reset timing mode. If no ground exists at pin D, the input voltage at pin A is greater than 18 0 VDC, and ground is removed from pin E for 200 microseconds or longer and then reapplied, pin C shall provide a ground path in accordance with 3 4 1.1 in a timing mode for 5.0 ± 0.5 minutes, after which it shall be disconnected from ground, and current shall be not greater than 0 005 ampere through any pin. If the removal of ground at pin E is accompanied by contact bounce, such that the ground is removed and reapplied more than once during the switch contact chatter period, the BRCU shall respond by providing a timed ground path.

3 4 1 4 Loss of input voltage If no ground exists at pin D, pin E is grounded, and input voltage of greater than 18 0 VDC is removed from pin A for 60 milliseconds or longer and then reapplied, pin C shall be disconnected from ground, and current shall be not greater than 0 005 ampere through any pin

3.4.1.5 Switching time The BRCU shall perform its intended switching function within 25 milliseconds of command after input voltage within the range of 18 to 33 VDC is applied The BRCU shall be capable of meeting this requirement with inductive or resistive loads from zero to 0 75 ampere at pin C Inductive DC test loads shall be in accordance with MIL-I-81023 for type II inductors

3 4 1.6 Ground sensing When ground is applied to pins D and E, a current of 0 010 amperes or greater shall flow through pin D When ground is applied to pin E, but no ground is applied to pin D, a voltage of 6 0 VDC or greater shall be present at pin D The values specified shall apply when the input operating voltage at pin A is 18 0 VDC or greater

3 4 1 7 Transient voltage immunity The BRCU shall be capable of maintaining the ground path from pin C to pin E while in the timed mode following power interruptions from pin A to pin B of not greater than 5 0 milliseconds Interruptions of this duration shall not cause the BRCU to initiate a new timing cycle

3 4.2 Electrical performance in alternate wiring configuration When connected as on figure 3, with the input voltage in the range specified by 3 3 8 applied from pin A to pin B, and a relay coil drawing as much as 0 75 ampere through pin C, the BRCU shall meet the requirements of 3 4 2 1 through 3 4 2.3

3 4 2 1 Ground path in alternate wiring configuration If a ground path is available at pins D and E through closure of the left DC relay contacts or through closure of the auxiliary power unit (APU) switch in combination with the battery switch or the BRCU reset switch, the BRCU shall provide a continuous ground path from pin C to pin E with a voltage drop not greater than 100 millivolts The ground path shall be free of abnormal interruptions and transients If a ground path exists at pin D but not at pin E, no ground path shall be available at pin C

3 4 2 2 Timing mode in alternate wiring configuration With the battery switch off, the BRCU reset switch off, the left DC relay contacts off, and the APU switch off, no ground path shall be available at



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pin C If the timing mode is initiated by a path provided from pin E to ground through the battery switch or the BRCU reset switch, the BRCU shall provide a timed ground path for 5 0 ± 0 5 minutes The ground path shall be free of abnormal interruptions and transients Following the timing period, the ground path from pin C to pin E shall be disconnected, and current shall be not greater than 0 005 ampere through any pin.

3 4 2.3 Reset timing period in alternate wiring configuration If, while providing a timed ground path at pin C as in 3.4 2.1, the ground path at pin E is interrupted for as little as 200 microseconds and then re-established (as when the battery switch is turned off and then back on) the BRCU shall provide a timed ground path for 5 0 ± 0 5 minutes from the time of the interruption The ground path shall be free of abnormal interruptions and transients If the interruption at pin E is accompanied by contact bounce, the BRCU shall respond to this as though it were a single interruption in the ground path and shall provide the timed ground path as specified, so long as a continuous interruption of at least 200 microseconds had occurred If the battery switch is on, the BRCU reset switch is off, the left DC relay is off, and the APU switch is turned from on to off, the BRCU shall provide a timed ground path for 5 0 ± 0 5 minutes after the APU switch is turned off.

3 4.3 Operating position The BRCU shall be capable of operating to the specification requirements of 3 4 through 3 4 2 3 in any orientation and attitude

3.5 Environmental When subjected to the environmental conditions listed in table I, the BRCU shall show none of the following

a Dimensional distortion beyond specified limits or cracking of cases, covers, or mounting brackets

b. Breakdown of insulation, loosening or stripping of metal plating or protective coatings from any component part, or corrosion of metal parts

c. Mechanical failure of any part

d Fungal growth

e Deterioration of BRCU identification markings

f. Degradation of electrical performance beyond specified limits under 3 4

g Abnormal interruptions and transients on the output.



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TABLE I. Environmental conditions

| Environmental condition | Test paragraph |
|---|---|
| Acceleration | 4 8 10 1 |
| Dust | 4 8 10.2 |
| Salt fog | 4.8.10.3 |
| Temperature-altitude | 4.8 10 4 |
| Mechanical shock | 4 8 10 5 |
| Fungus | 4 8 10 6 |
| Humidity | 4.8 10 7 |
| Sine vibration | 4 8 10 8 |
| Random vibration | 4 8 10 9 |
| Explosive atmosphere | 4 8 10 10 |

3.6 Electromagnetic compatibility With the input and load source voltage in the range specified by 3 3 8, the BRCU shall meet the following electromagnetic compatibility requirements of MIL-STD-461, as specified for category A1b of equipment class A1: CE01, CE03, CE07, CS01, CS02, CS06, RE01, RE02, RS01, RS02,and RS03 except that the RS03 frequencies and field intensities shall be as follows

RS-03 Test levels

| Frequency range | Electric field (volts/meter) |
|---|---|
| 14 kilohertz - 200 megahertz | 20 |
| > 200 megahertz - 450 megahertz | 65 |
| > 450 megahertz - 1 0 gigahertz | 20 |
| > 1.0 gigahertz - 12 0 gigahertz | 100 |
| > 12 0 gigahertz - 40 0 gigahertz* | 20* |

*Except 100 volts/meter at 16 5 and 33 0 gigahertz.

3 7 Marking. Markings shall be in accordance with MIL-STD-130

3 8 Nameplate Each BRCU shall contain one or more appropriately sized type A, B, or H nameplates in accordance with MIL-P-15024 The nameplate(s) shall be attached securely to the unit. Alternate nameplates are acceptable so long as the legibility and durability requirements of MIL-B-85584B(AS) are complied with Minimum lettering height shall be 080 inch The manufacturer's name or trademark shall be in letters not larger than other letters appearing on the nameplate(s) The date of manufacture and the warranty expiration date may be optionally printed on the label as calendar year



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quarters If this technique is used, the two dates shall be separated by 23 quarters The nameplate(s) shall contain the following information

Aircraft Battery Relay Control Unit
Input 28 volts DC, 0.050 ampere DC (Max)
MIL P/N D85584-1
5945-01-207-9035
Manufacturer's part number and revision
Contract or order number
Serial number
Date manufactured
Manufacturer's name or trademark
Inspector's stamp
United States
Warranty expiration date
THIS UNIT IS NONREPAIRABLE
DO NOT ATTEMPT TO DISASSEMBLE

3 9 Workmanship BRCUs shall be manufactured and assembled in accordance with MIL-STD-454, requirement 9

3 10 Soldering Soldering shall comply with MIL-STD-454, requirement 5

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements

4 1 1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract

4 1 2 Sampling inspection Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material

4 2 Classification of inspections The inspection requirements specified herein are classified as follows

a Qualification inspection (see 4 4)



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b Quality conformance inspection (see 4 5)

c Qualification verification inspection (see 4 6)

4 3 Inspection conditions Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.7. The inspection system requirements shall be in accordance with MIL-I-45208.

4 4 Qualification inspection. In accordance with SD-6, the Government shall perform qualification inspection on samples of BRCUs furnished by the supplier Qualification tests shall consist of the examinations, inspections, and tests specified in table II and shall be performed in the order shown.

4 4 1 Qualification samples. The submission for qualification shall consist of three test samples which shall be produced to the maximum extent practicable with equipment and procedures normally used in production and shall be identified as specified (see 3 7 and 3 8) Tests shall be conducted as specified in table II under any one of the sample number columns selected at the discretion of the qualifying activity. Any additional information required by the letter of authorization specified by SD-6 shall be supplied (see 6.3)

4 4.2 Materials certification. The manufacturer shall prepare a report to certify that the materials and finishes used in fabricating BRCUs are in accordance with the requirements of 3 2 and the applicable referenced specifications This certifying report shall include verifying data and shall be submitted with the qualification test samples



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TABLE II Schedule of qualification and qualification verification tests

| Test Title | BRCU Sample Number | | | Requirement paragraph | Test Paragraph |
|---|---|---|---|---|---|
| | 1 | 2 | 3 | | |
| Marking | x | x | x | 3.7 | 4.8 1 |
| Nameplate | x | x | x | 3.8 | 4.8.1 |
| Visual and mechanical inspection | x | x | x | 3.2 and 3 3 | 4 8 1 |
| Input voltage | x | x | x | 3.3.8 | 4.8 2 |
| Input current | x | x | x | 3.3.9 | 4.8.3 |
| Input power characteristics | | x | | 3.3 10 | 4.8 4 |
| Electrical connection sequence | x | x | x | 3.3 13 | 4.8.5 |
| Insulation resistance | x | x | x | 3 3 14 | 4 8.6 |
| Electrical performance | | | | 3 4 1 | 4 8.7 |
| Ground path | x | x | x | 3.4 1 1 | 4.8 7.1 |
| Timing period | x | x | x | 3 4 1.2 | 4.8 7.2 |
| Reset timing mode | x | x | x | 3 4 1 3 | 4 8 7 3 |
| Loss of input voltage | x | x | x | 3 4 1 4 | 4 8 7 4 |
| Switching time | x | | | 3 4 1 5 | 4 8 7.5 |
| Ground sensing | | | x | 3 4 1 6 | 4 8 7 6 |
| Transient voltage immunity | | | x | 3 4 1 7 | 4 8 7 7 |
| Electrical performance in alternate wiring configuration | | | | 3 4 2 | 4 8 8 |
| Ground path in alternate wiring configuration | x | x | x | 3 4 2 1 | 4.8 8 1 |
| Timing mode in alternate wiring configuration | x | x | x | 3 4 2 2 | 4 8 8 2 |
| Reset timing mode in alternate wiring configuration | x | x | x | 3 4 2 3 | 4 8 8 3 |
| Operating position | x | | | 3 4.3 | 4.8 9 |
| Acceleration | x | | | 3 5 | 4 8 10 1 |
| Dust | x | | | 3 5 | 4 8 10 2 |
| Salt fog | x | | | 3 5 | 4 8 10 3 |
| Temperature-altitude | | x | | 3 5 | 4 8 10 4 |
| Mechanical shock | | x | | 3 5 | 4 8.10 5 |
| Fungus | | x | | 3.5 | 4 8 10 6 |
| Humidity | | | x | 3 5 | 4.8 10 7 |
| Sine vibration | x | | | 3 5 | 4 8 10 8 |
| Random vibration | | | x | 3 5 | 4 8 10 9 |
| Explosive atmosphere | | | x | 3 5 | 4 8 10 10 |
| Electromagnetic compatibility | | | x | 3 6 | 4 8.10 11 |
| Disassembly and inspection | x | x | x | 3 2, 3 9, 3 10 | 4 8 11 |



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4 5 Quality conformance inspection The manufacturer shall perform the quality conformance inspections of table III on each BRCU prior to delivery Data from these inspections shall be recorded and available for examination by authorized Government representatives

TABLE III Quality conformance inspection

| Test title | Group | Requirement paragraph | Test paragraph |
|---|---|---|---|
| Insulation resistance | A | 3.3 14 | 4.8 6 |
| Electrical performance | A | 3 4.1 | 4 8.7 through 4 8.7.7 |
| Visual and mechanical inspection | A | 3.2 and 3.3 | 4 8.1 |
| Marking | A | 3 7 | 4.8.1 |
| Nameplate | A | 3.8 | 4.8.1 |

4.5 1 Noncompliance If a sample fails to pass quality conformance inspection, the manufacturer shall notify the procuring activity of such failure and take corrective action on the materials or processes, or both, as warranted The manufacturers shall also correct all units of the product manufactured under essentially the same materials and processes and which are considered subject to the same failure Acceptance and shipment of the product shall be discontinued until corrective action acceptable to the procuring activity has been taken After the corrective action has been taken, quality conformance inspection shall be repeated on additional sample units (either all tests and examinations or the test which the original sample failed, at the option of the procuring activity) Final acceptance and shipment shall be withheld until the quality conformance inspection has shown that the corrective action was successful

4 6 Qualification verification inspection Each BRCU of the same part number delivered as a qualified item under this specification shall be numbered sequentially in the order that it is delivered. To meet the delivery requirement for the first 50 BRCUs, 51 shall successfully complete the required quality conformance inspections Of the 51 units, 50 shall be shipped as directed by the applicable contract(s), and 1 BRCU, selected at random by the Government inspector, shall be forwarded to the qualifying activity (see 6 3). To meet the delivery requirement for each succeeding 150 BRCUs, 151 BRCUs shall successfully complete the required quality conformance inspections Of the 151 units, 150 shall be shipped as directed by the applicable contract(s), and 1 BRCU selected at random shall be forwarded as above The span of serial numbers and the contract numbers of the production units from which each verification sample was selected shall be recorded and included with the sample. The sample shall be shipped no later than the final unit in the production group from which selected The units forwarded to the qualifying activity shall undergo qualification verification inspection Tests shall be conducted as specified in table II under any one of the sample number columns selected at the discretion of the qualifying activity. All qualification verification samples shall become Government property when accepted by the Government inspector for shipment to the qualifying activity.

4 6 1 Rejection and retest of qualification and qualification verification samples Samples which have been rejected or returned to the manufacturer for any reason during qualification or qualification verification tests may be reworked or have parts replaced to correct defects. Before resubmitting the samples, the manufacturer shall submit to the qualifying activity a written report stating the full particulars concerning the rejection and the corrective action taken The cost of retests shall be borne by the manufacturer Testing shall not be resumed until such a report is received. The manufacturer's unit may be disqualified at the discretion of the qualifying activity if the manufacturer



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a Does not correct and return samples rejected during qualification testing within 5 weeks after receipt from the qualifying activity, or

b Does not submit within 2 weeks a program for the correction of deficiencies that is acceptable to the qualifying activity.

4 6.2 Retention of qualification In order to retain qualification of a product approved for listing on the Qualified Products List (QPL), the manufacturer shall certify to the qualifying activity that the product complies with the requirements of this specification The time of periodic certification shall be in 2-year intervals from the date of original qualification The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine whether the product continues to meet any or all of the specification requirements

4 7 Test conditions Tests shall be conducted under the conditions specified in 4 7.1 through 4 7.6.

4 7 1 Standard ambient conditions Unless otherwise specified, tests shall be performed under the following ambient atmospheric conditions

a Temperature 23 ± 5°C

b Relative humidity. 50 ± 30 percent

c Atmospheric pressure 725 + 50, -75 mm of mercury

4 7 2 Measurements and tolerances Measurements shall be made with instruments and test equipment in accordance with MIL-E-81910 Tolerances of nonstandard ambient conditions shall be in accordance with MIL-E-81910.

4 7.3 Ambient temperature measurements When tests are conducted at reduced or elevated temperatures, the ambient temperature shall be determined by measuring the air temperature at several points surrounding, but not above, the unit and averaging the results

4 7.4 Airflow. For all temperature-altitude tests below 6 1 kilometers, the airflow over the battery relay control unit within the test chamber shall be adjusted to approximate natural convection conditions. The airflow shall be not greater than 30.5 meters per minute while maintaining the required ambient temperature within specified tolerance. For tests above 6 1 kilometers altitude, the airflow may exceed 30.5 meters per minute but shall be adjusted to the minimum velocity consistent with maintaining the required ambient temperature within specified tolerance

4 7.5 Position Unless otherwise specified, the units shall be mounted base down

4 7.6 Input voltage Nominal or other specified input voltage shall be applied as indicated herein Nominal voltage shall be 28 VDC

4 7 7 Inductive load Inductive DC loads required during testing shall be in accordance with MIL-I-81023, type II inductor

4 8 Methods of inspection The following tests and inspections shall be performed



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4.8.1 Visual and mechanical inspection Inspect the BRCU to determine compliance with the requirements of this specification with respect to materials, parts, workmanship, dimensions, weight, markings, and connector orientation.

4 8 2 Input voltage

4.8.2.1 Voltage limits for initiating the ground path Test the BRCU to verify its capability of initiating the ground path in accordance with 3.4.1.1 for input voltages of 33 0 VDC, nominal, and 18.0 VDC. The BRCU is not required to initiate the ground path when input voltage is below 18.0 VDC.

4.8.2.2 Voltage limits for maintaining the ground path Test the BRCU to verify its capability of maintaining the ground path in accordance with 3 4 1 1 for input voltages of 33 0 VDC, nominal, and 10.0 VDC.

4.8 2.3 Voltage limits for terminating the ground path Test the BRCU to verify its capability of terminating the ground path in accordance with 3.4 1.1 for input voltages of 33.0 VDC, nominal, and 10.0 VDC

4 8 3 Input current Apply input voltage to pin A with no load source voltage shall be applied to pin C Apply ground continuously to pin B and as necessary to pins D and E Monitor the input operating current at pin A for all functional operations of 3 4 1 1, 3 4 1 2, and 3 4 1 3 to verify that it does not exceed 0 050 ampere, as specified in 3.3.9.

4 8 4 Input power characteristics Test the BRCU to determine conformance to the requirements of 3 3 10 Apply the following transients to the BRCU pin A while the unit is operating in accordance with 3 4 1 1 and 3.4.1.3

a. 50 VDC for 12 5 milliseconds

b 46 VDC for 25 milliseconds

c 40 VDC for 47 5 milliseconds

The unit shall not be damaged during this test. The unit shall function in accordance with 3 4 1 1 and 3 4 1 3 both during and following application of the transients.

4.8.5 Electrical connection sequence. Connect input voltage to pin A, ground to pin B, a 0.75 ampere inductive DC load source to pin C, and ground as necessary to pins D and E. Operate the BRCU in each of the modes of 3.4.1 through 3.4 1 7. Remove and then reapply the electrical input at each pin taken in random sequence during operation in each of the modes. Verify that the BRCU meets the requirements of 3.3.13 and 3 4 1 through 3.4.1.7.

4.8 6 Insulation resistance. Measure the insulation resistance between all pins and the unpainted electrical grounding area at the case mounting holes in accordance with MIL-STD-202, method 302, test condition B, with the ambient relative humidity at 30 to 50 percent Verify that the insulation resistance conforms to 3 3.14.



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4 8 7 Electrical performance With the BRCU connected as shown on figure 2, evaluate electrical operation as specified in 4.8.7.1 through 4 8.7 7 Unless otherwise specified, apply input voltage from pin A to pin B, and a 0 75 ampere inductive load to pin C. Use an appropriate mating plug connector conforming to DOD-C-38999/46 (such as D38999/46WB5SN) Test the BRCU over the full withstanding, initiating, and maintaining input voltage ranges of 3 3 8

4 8.7 1 Ground path Ground pins D and E Verify that pin C provides a path to ground in compliance with 3.4 1 1. Remove the ground from pin E Verify that pin C is disconnected from ground and that current is not greater than 10 microamperes through any connector pin

4.8.7.2 Timing period. Ground pins D and E Remove the ground at pin D and begin timing. Verify that pin C provides a ground path in accordance with 3 4 1.2 for 5.0 ± 0.5 minutes After 5.0 ± 0.5 minutes, verify that pin C is disconnected from ground and that current is not greater than 0.005 ampere through any pin Reapply ground to pin D, and verify that pin C provides a ground path in accordance with 3 4 1 1

4 8 7.3 Reset timing period. Ground pins D and E to establish a ground path through pin C. Remove the ground from pin D Momentarily remove the ground from pin E for not less than 200 microseconds Verify that pin C provides a ground path in accordance with 3 4 1 3 for 5 0 ± 0 5 minutes from the time of the ground interruption at pin E and that, after the timing period, there is no current greater than 0 005 ampere through any connector pin

4 8 7 4 Loss of input power Provide a ground path for pin C by grounding pin E, but not pin D Momentarily remove the input voltage from pin A for not less than 60 milliseconds. Verify that pin C is disconnected from ground in accordance with 3 4 1 4, and current is not greater than 0.005 ampere through any connector pin

4 8 7 5 Switching time With pin D ungrounded, ground pin E Verify that in accordance with 3 4 1.5 a ground path is provided by pin C within 25 milliseconds

4 8 7 6 Ground sensing Apply 18 VDC to pin A, and ground pins D and E Verify that the current through pin D is not less than 10 milliamperes in accordance with 3 4 1 6 Remove ground from pin D and verify that the voltage at pin D is not less than 6 0 volts

4 8 7.7 Transient voltage immunity Apply 18 to 33 VDC from pin A to pin B while pins D and E are ungrounded Connect pin E to ground to establish a ground path and timing mode at pin C Momentarily remove input voltage for not less than 5 0 milliseconds In accordance with 3.4.1.7, verify that the ground path through pin C is maintained during and after this power interruption, and a new timing mode is not initiated as a result of this interruption

4.8 8 Electrical performance in alternate wiring configuration With the BRCU connected as shown on figure 3, confirm conformance to the requirements of 3 4 2 1 through 3 4 2 3 as specified in 4 8 8 1 through 4 8 8 3 Apply input voltage within the ranges specified in 3 3 8 from pin A to pin B. Apply a 0 75 ampere inductive load to pin C Use an appropriate mating plug connector conforming to DOD-C-38999/46 (such as D38999/46WB5SN).

4 8 8 1 Ground path in alternate wiring configuration With the battery switch turned off, the BRCU momentary (MOM) reset switch off, the left DC relay contacts off, and the APU (APU) switch turned off, verify that verify the output of the battery control relay (BCR) is off With the APU switch turned on, verify



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that the output of the battery control relay remains off. Ground pin E momentarily, closing the BRCU reset switch Verify that the output of the battery control relay is on and remains on continuously. Next, turn the APU switch off. Verify that the output of the battery control relay turns off in 5 0 ± 0.5 minutes Upon cessation of this timing mode, ground pins D and E by closing the left DC relay contacts. Verify that the output of the battery control relay is on continuously through a ground path at pin C.

4.8.8.2 Timing mode in alternate wiring configuration With the battery switch off, the BRCU MOM reset switch off, the left DC relay contacts open, and the APU switch off, verify that the output of the battery control relay is off Turn the battery switch on Verify that a ground path is provided at pin C and that the output of the battery control relay is on and remains on for 5 0 ± 0.5 minutes. After this timing mode has been completed, verify that pin C is disconnected from ground and that current is not greater than 0.005 ampere through any pin. Turn the battery switch off. Verify that the output of the battery control relay remains off Turn the BRCU reset switch on. Verify that a ground path is provided at pin C and that the output of the battery control relay is on Release the BRCU reset switch. Verify that the ground path remains at pin C and the output of the battery control relay remains on Verify that following period of 5.0 ± 0.5 minutes pin C is disconnected from ground and there is no current greater than 0.005 ampere through any connector pin.

4 8 8 3 Reset timing mode in alternate wiring configuration With the battery switch off, the BRCU momentary reset switch off, the left DC relay contacts open, and the APU switch turned off, verify that the output of the battery control relay is not energized Turn the battery switch on Verify that a ground path is provided at pin C and that the output of the battery control relay is on After the output of the battery control relay has been on for 2 minutes, momentarily turn the battery switch off for not less than 200 microseconds Verify that the output of the battery control relay stays on while the battery switch is momentarily turned off and on, and remains on for 5.0 ± 0 5 minutes after the battery switch is turned back on. After this timing period has been completed, turn the APU switch on. Verify that the output of the battery control relay is on After the output of the battery control relay has been on for 2 minutes, turn the APU switch off Verify that the output of the battery control relay remains on for 5.0 ± 0.5 minutes after the APU switch was turned on

4 8.9 Operating position Test the BRCU to the requirements of 3 4, alternately placing it on each of its six surfaces Confirm that the BRCU meets the performance requirements of 3 4 through 3.4 2 3 in all orientations

4 8.10 Environmental.

4 8 10 1 Acceleration. Test the BRCU in accordance with the acceleration test requirements of MIL-E-81910, except with an acceleration level of 15 *g* During procedure II, operate the BRCU at its rated load with nominal input voltage applied Monitor the current through each connector pin Verify that the output is free of abnormal interruptions and transients. At the conclusion of the test, verify that the BRCU complies with 3.5. Test the BRCU in accordance with 4 8.7 through 4.8.8.3 to verify that the BRCU meets the requirements of 3.4 1 through 3.4.2.3

4 8.10 2 Dust Test the BRCU in accordance with the dust test requirements of MIL-E-81910, except do not apply input voltage during the test At the conclusion of the test, verify that the BRCU complies with 3 5 Test the BRCU in accordance with 4 8 7 through 4 8 8 3 to verify that the BRCU meets the requirements of 3 4.1 through 3.4.2 3



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4 8.10 3 Salt fog Test the BRCU in accordance with the salt fog test requirements of MIL-E-81910 Following the test, allow the unit to stand at room ambient conditions for 24 hours, and then test the BRCU in accordance with 4.8.7 through 4 8.8.3 to verify that the BRCU meets the requirements of 3 4 1 through 3 4.2 3 After an additional 48-hour stand at room ambient conditions, examine the BRCU for compliance with 3.5, and test the BRCU in accordance with 4 8 7 through 4 8 8 3 to verify that the BRCU meets the requirements of 3.4 1 through 3 4.2 3.

4.8.10 4 Temperature-altitude. Test the BRCU in accordance with the temperature-altitude test requirements of MIL-E-81910 for self cooled equipment, except that the maximum and minimum test temperatures shall be 71°C and -55°C, respectively The maximum test altitude shall be 20 kilometers Evaluate the BRCU electrically at 4-kilometer intervals in accordance with 4 8.7 through 4.8.8.3 to verify that the BRCU meets the requirements of 3 4 1 through 3.4 2.3 At the conclusion of the test, verify that the BRCU complies with 3 5 Test the BRCU in accordance with 4.8 7 through 4.8 8.3 to verify that the BRCU meets the requirements of 3 4 1 through 3 4 2 3

4 8.10 5 Mechanical shock Test the BRCU in accordance with the mechanical shock test requirements of MIL-E-81910, except do not perform the equipment crash hazard test, procedure III of MIL-STD-810 At the conclusion of the test, verify that the BRCU complies with 3 5 Test the BRCU in accordance with 4 8 7 through 4 8 8 3 to verify that the BRCU meets the requirements of 3 4 1 through 3 4 2.3

4 8 10 6 Fungus Test the BRCU in accordance with the fungus test requirements of MIL-E-81910 At the conclusion of the test, verify that the BRCU complies with 3 5 Test the BRCU in accordance with 4 8 7 through 4.8 8.3 to verify that the BRCU meets the requirements of 3 4.1 through 3 4.2 3.

4.8 10 7 Humidity Test the BRCU in accordance with the humidity test requirements of MIL-E-81910 Operate the BRCU at its rated load with nominal input voltage applied while the BRCU is at 30°C during each cycle of the humidity test. Monitor the current through each connector pin Verify that the output is free of abnormal interruptions and transients At the conclusion of the test, verify that the BRCU complies with 3 5 Test the BRCU in accordance with 4 8 7 through 4 8 8 3 to verify that the BRCU meets the requirements of 3 4 1 through 3 4.2 3

4.8.10 8 Sine vibration. Test the BRCU in accordance with the sine vibration test of MIL-E-81910 for non-engine-mounted components, except use the following sine vibration test displacements, acceleration levels, and crossover frequencies

| Frequency | Test Level |
|---|---|
| 5 Hz through 14 Hz | 5 08 mm double amplitude (DA) |
| > 14 Hz through 33 Hz | 2 *g* |
| > 33 Hz through 74 Hz | 0 9144 mm DA |
| > 74 Hz through 2000 Hz | 10 *g* |

During each axis of vibration, and at the conclusion of the test, operate the BRCU at its rated load with nominal input voltage applied. Monitor the current through each connector pin Verify that the output is free of abnormal interruptions and transients At the conclusion of the test, verify that the BRCU complies with 3 5 Test the BRCU in accordance with 4 8 7 through 4 8 8 3 to verify that the BRCU meets the requirements of 3 4 1 through 3 4 2 3



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4.8.10.9 Random vibration Test the BRCU in accordance with the random vibration test of MIL-E-81910 for non-engine mounted components, except subject the BRCU to only the functional level of random vibration for the entire test time of 2 hours per axis During each axis of vibration, and at the conclusion of the test, operate the BRCU at its rated load with nominal input voltage applied. Monitor the current through each connector pin Verify that the output is free of abnormal interruptions and transients. At the conclusion of the test, verify that the BRCU complies with 3.5. Test the BRCU in accordance with 4.8 7 through 4 8.8.3 to verify that the BRCU meets the requirements of 3.4.1 through 3 4.2.3.

4 8.10.10 Explosive atmosphere. Subject the BRCU to the explosive atmosphere test of MIL-STD-810, method 511, procedure I. Test up to a chamber temperature of not greater than 71°C and a test altitude of not greater than 20 kilometers. During the test, operate the BRCU at its rated load with nominal input voltage applied. At the conclusion of the test, verify that the BRCU complies with 3.5. Test the BRCU in accordance with 4.8.7 through 4 8.8.3 to verify that the BRCU meets the requirements of 3.4.1 through 3.4.2.3

4.8 10.11 Electromagnetic compatibility. Test the BRCU in accordance with the applicable test methods and procedures of MIL-STD-462 to determine compliance with the requirements of 3.6. During the test, operate the BRCU at its rated load with nominal input voltage applied. Monitor the current through each connector pin Verify that the output is free of abnormal interruptions and transients At the conclusion of the test, verify that the BRCU complies with 3 5 Test the BRCU in accordance with 4 8 7 through 4 8 8 3 to verify that the BRCU meets the requirements of 3 4 1 through 3.4.2.3

4 8 11 Disassembly and inspection At the conclusion of all tests, disassemble and inspect each sample for evidence of excessive deterioration or defects which could prevent the BRCUs from performing their intended functions Inspect for compliance with the materials requirements of 3 3, for degradation caused by environmental testing as specified in 3.5, and for compliance with the workmanship requirements of 3.9. The presence of such evidence shall be cause for rejection.

4.8.12 Packaging inspection The preservation, packing, and marking shall be inspected to verify conformance to the requirements of Section 5.

## 5 PACKAGING

5.1 Preservation Preservation shall be in accordance with MIL-STD-2073-1, level A or C, as specified in the contract or purchase order (see 6.2)

5.1.1 Level A Unless otherwise specified in the contract or purchase order, level A preservation shall be in accordance with MIL-P-116, method II, without preservation compounds, using metal reusable containers

5.1.2 Level C. For level C, the BRCU shall be preserved in accordance with the manufacturer's commercial practice

5 2 Packing Packing shall be in accordance with MIL-STD-2073-1, level A, B, or C as specified in the contract or purchase order

5 3 Marking Interior and exterior containers shall be marked in accordance with MIL-STD-2073-1 and MIL-STD-129



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## 6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6 1 Intended use The BRCU is intended for use in aircraft to prevent inadvertent discharge of the aircraft storage battery after the aircraft is shut down The unit automatically controls the aircraft battery relay, which will electrically disconnect the aircraft battery from the electrical bus after sufficient time has elapsed for normal use of the battery once the aircraft is shut down on the ground

6.2 Acquisition requirements Acquisition documents must specify the following

a Title, number, and date of the specification.

b Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c Military part number (see 3 8 and figure 1)

d. Qualification verification samples (see 4 6)

e Required level of preservation and packing (see 5 1 and 5 2)

f. Type of container (see 5 1 1)

6 3 Qualification With respect to products requiring qualification, awards will be made only for products which are, at the time of award of the contract, qualified for inclusion in Qualified Products List QPL No. 85584 whether or not such products have actually been so listed by that date The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification The activity responsible for the Qualified Products List is the Naval Air Systems Command, AIR-53624, Washington, DC 20361-5360, however, information pertaining to qualification of products may be obtained from Crane Division, Naval Surface Warfare Center, Code 6092, Building 2949, Crane, IN 47522-5060

6 4 Engineering changes to a qualified item Once a manufacturer's part number has been placed on the Qualified Products List, the manufacturer is responsible for any changes to the BRCU which affect its physical or performance characteristics No changes may be made to a qualified unit without the specific written approval of the qualifying agency The manufacturer should allow sufficient lead time for the qualifying agency to repeat the qualification tests if necessary

6 5 Installation The BRCU will be installed in any position in the aircraft in accordance with MIL-E-7080

6 5 1 Installation clearances Clearance should be provided for installation of BRCU mounting hardware and for connection with mating plug connector D38999/46WB5SN



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6 5.2 System integration. Airframe manufacturers are cautioned that the BRCU may not operate satisfactorily in circuit configurations different from those shown on figures 2 and 3. No testing in other configurations has been performed

6 6 Design guides The manufacturer should consider the following design guidelines, which are based on actual experience, when a BRCU is to be developed and proposed in accordance with this specification

6 6.1 Environmental stress screening Environmental stress screening of electronic parts should be used to meet the warranty requirements as specified in the contract.

6 6.2 Semiconductors and electron tubes Semiconductors should be chosen and applied as outlined in MIL-STD-454, requirement 30 Electron tubes should not be used

6.6.3 Internal electrical connections. Internal connections should be made in accordance with MIL-STD-454, requirements 1 and 69

6 6 4 Adjustable resistors Only sealed adjustable resistors conforming to MIL-R-39015 should be used

6 6 5 Microelectronic devices Microelectronic devices should be chosen and applied in accordance with MIL-STD-454, requirement 64 and should be listed in MIL-STD-1562.

6 6.6 Capacitors Capacitors should be chosen and applied in accordance with MIL-STD-454, requirement 2 The use of wet slug tantalum capacitors is not recommended.

6 7 Manufacturing guides. The appropriate technical requirements contained in the following list of documents should be addressed and considered by the manufacturer where applicable

| Document | Title |
|---|---|
| MIL-STD-275 | Printed Wiring for Electrical Equipment |
| QQ-P-416 | Plating, Cadmium (Electrodeposited) |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |
| MIL-C-5541 | Chemical Conversion Coating on Aluminum and Aluminum Alloys |
| MIL-A-8625 | Anodic Coatings, for Aluminum and Aluminum Alloys |
| MIL-A-21180 | Aluminum Alloy Castings, High Strength |
| MIL-P-55110 | Printed Wiring Boards |
| WS-6536 | Procedures and Requirements for Preparation and Soldering of Electrical Connections |
| NAVMAT-P-9492 | Navy Manufacturing Screening Program |

6 8 Part or Identifying Number (PIN) The PIN to be used for the BRCUs acquired to this specification is created as follows: D85584-1



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6 9 Subject term (key word) listing

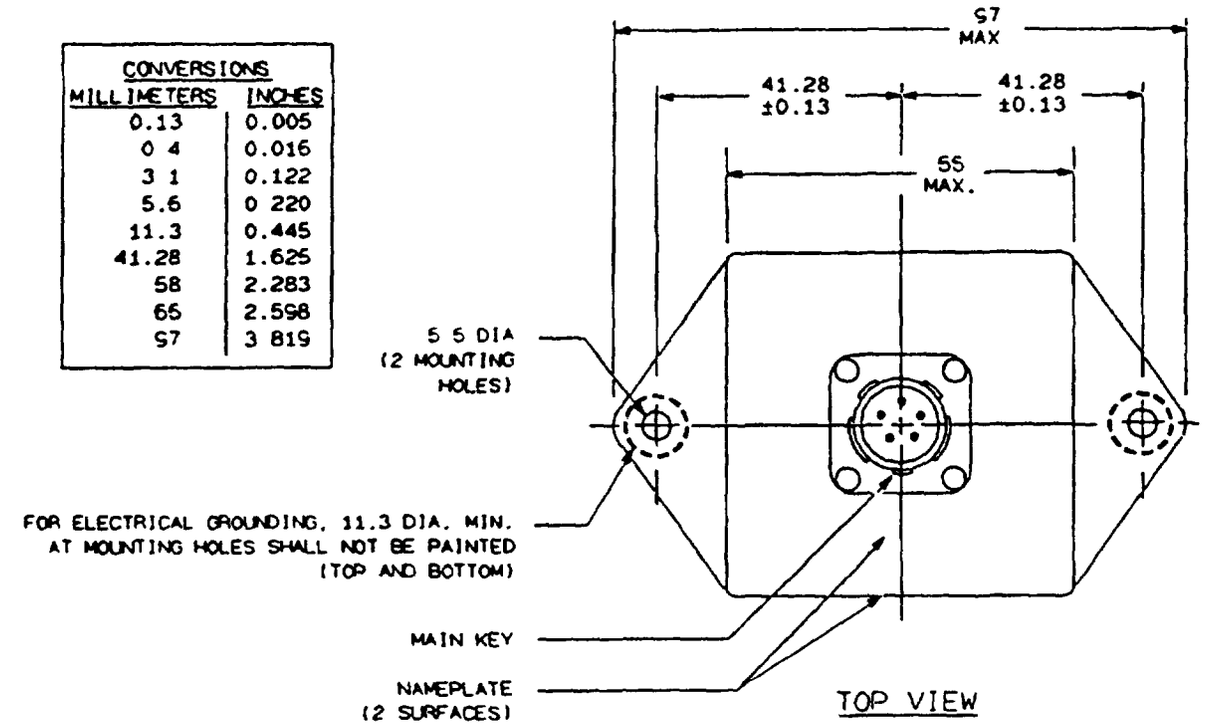
Electrical Bus
Connector
Convection cooling
Continuous duty
Electrical performance
Electromagnetic compatibility
Environmental testing requirements
Ground path
Ground sensing
Input current
Input power
Input voltage
Insulation resistance
Mounting
Operating position
Reset timing period
Switching time
Timing period
Transient voltage immunity
Unintentional discharge

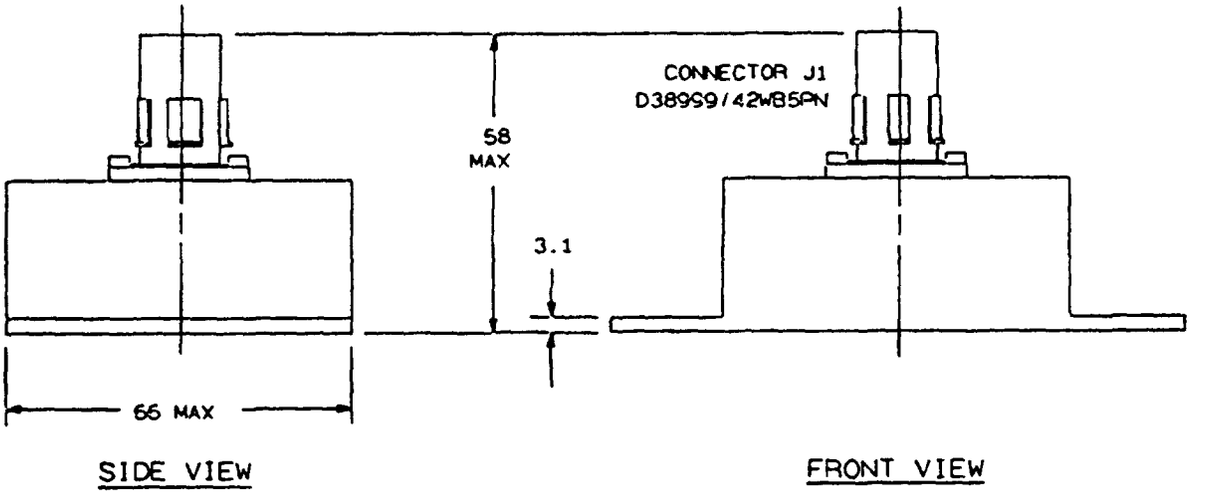
6.10 Changes from previous issue Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes



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| CONVERSIONS | |
|---|---|
| MILLIMETERS | INCHES |
| 0.13 | 0.005 |
| 0 4 | 0.016 |
| 3 1 | 0.122 |
| 5.6 | 0 220 |
| 11.3 | 0.445 |
| 41.28 | 1.625 |
| 58 | 2.283 |
| 66 | 2.598 |
| 97 | 3 819 |



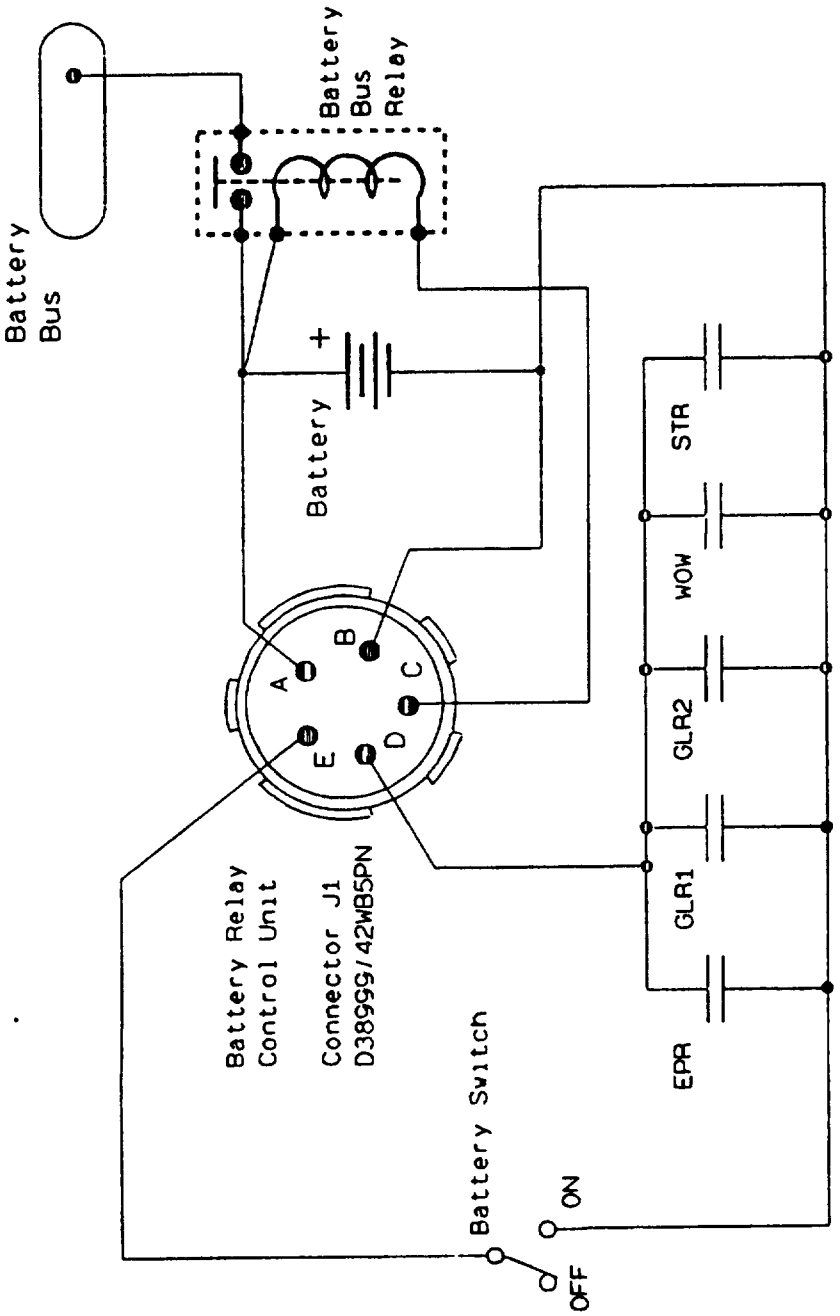

PART NUMBER D85584-1

NOTES
1. Dimensions are in millimeters
2. Tolerance is ±0.4 millimeters unless otherwise shown.
3. Inch-pound equivalents are given for information only.

FIGURE 1 Battery relay control unit dimensions



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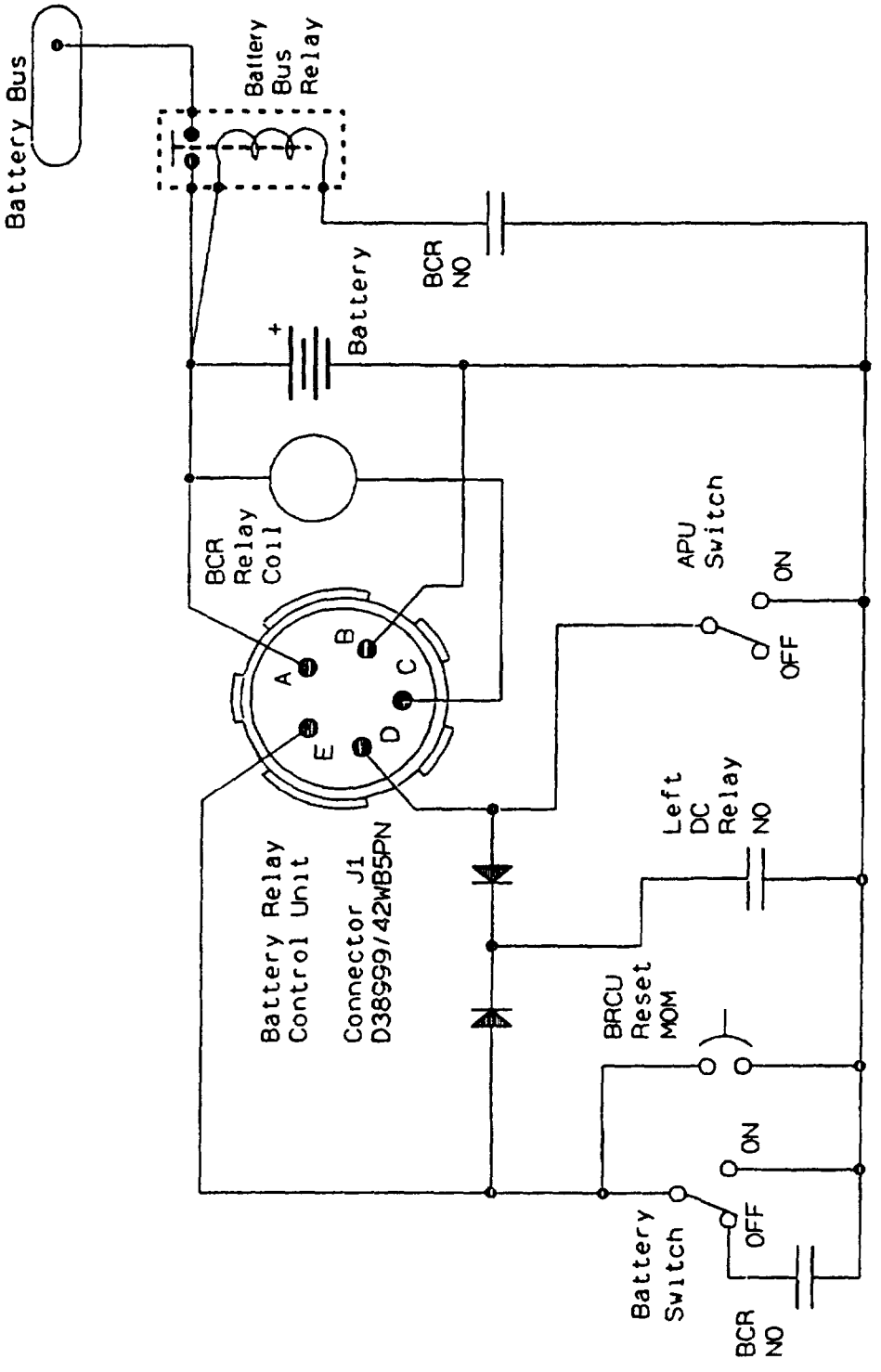



NOTES:
1 EPR - External power relay is closed when the aircraft is receiving external electrical power
2 GLR - Generator line relays are closed when aircraft generators are providing power.
3 WOW - Weight on wheels relays are closed when there is no weight on landing gear wheels
4 STR - Start relay closed during engine starting

FIGURE 2 Aircraft wiring connections to battery relay control unit connector.



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Notes:

1. BCR - Battery control relay controls contactor supplying power to aircraft electrical equipment
2. BRCU RESET - BRCU reset switch is located in the wheelwell of the aircraft.
3. LEFT DC RELAY - Left DC relay contacts are closed when the left D C generator is supplying power to the aircraft.
4. APU SWITCH - The APU Switch is closed during engine starting
5. NO - This designates normally open contacts when relay is not energized.

FIGURE 3 Battery relay control unit wiring connections, alternate configuration.



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